

Exploratory Data Analysis (EDA) & Modeling

- Box Plot and Linear Regression-

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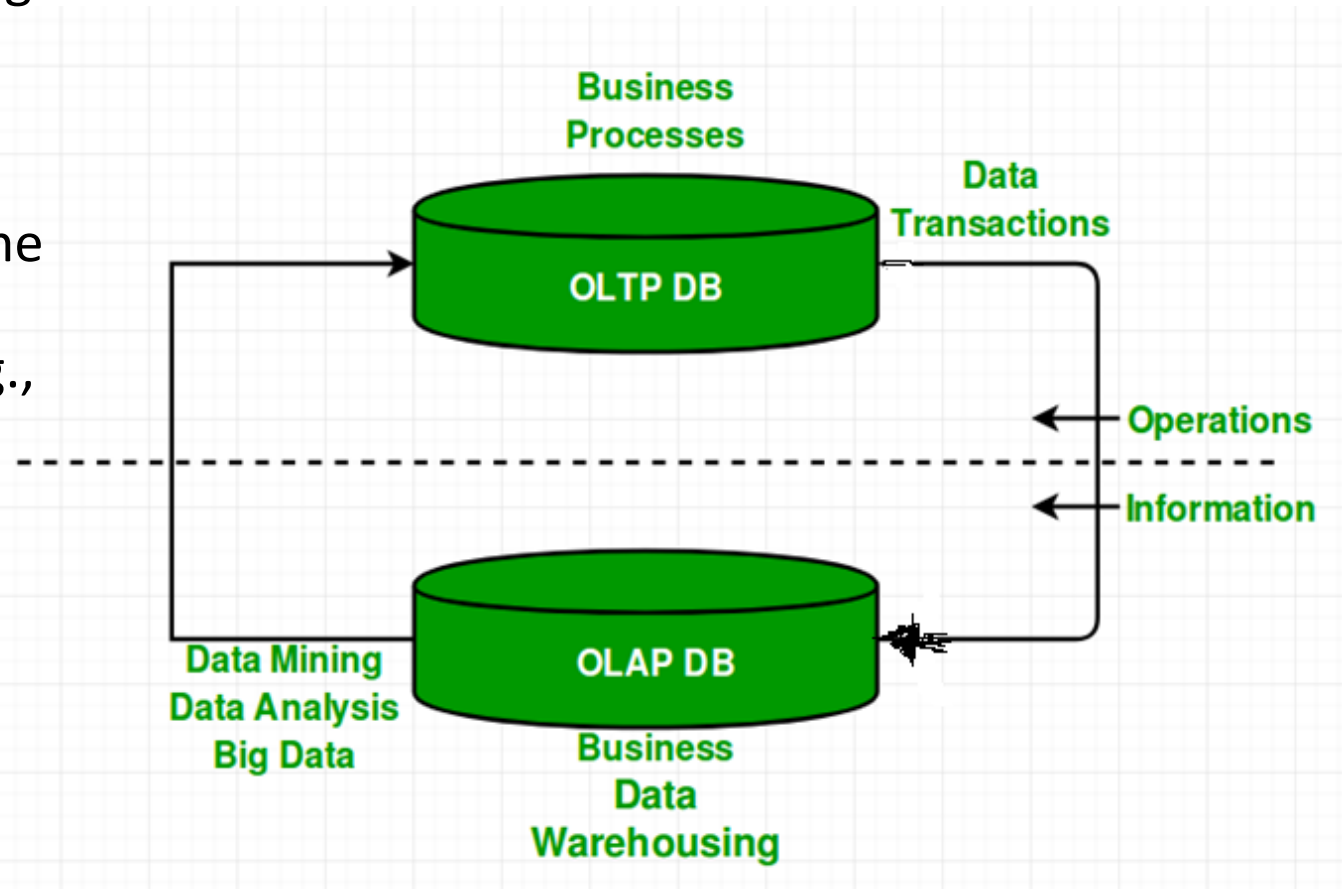
Source:

Beginning Data Science with R by M. A. Pathak, Springer International Publishing, 2014

An Introduction to Data Science by Jeffrey S. Saltz, Jeffrey M. Stanton, SAGE Publications, 2018

Evolution: data science (DS)

- One of the most important aspects of DS has to do with finding or recognizing (recognizing means that the patterns were not expected in advance) and evaluating the patterns.
- Evolving timeline
 - Relational databases (1970, later became DBMS)
 - OLTP (help answer some questions, e.g., online banking)
 - OLAP (data exploration and visualization; multidimensional databases, e.g., data warehouse)
 - Pattern recognitions; recommendation systems
 - Big data (with the need of DS, ML and DL)



EDA

Exploratory data analysis (EDA) is a data analysis technique that helps you understand your data by summarizing its main characteristics, often using statistical graphics and other data visualization methods. EDA is a non-parametric approach to data analysis, meaning that it does not make any assumptions about the underlying distribution of the data.

- Its purpose is **to understand the main characteristics** of the data
- E.g., a box plot is a type of EDA chart
- Linear regression is not EDA, but it is a type of data modeling
- EDA can be used to **identify patterns and trends in the data**, which can then be used to build a linear regression model
- EDA includes data visualization, statistical summaries, hypothesis testing

EDA

EDA is an iterative process that involves the following steps:

1. **Data cleaning and preparation:** The first step in EDA is to clean and prepare your data. This includes removing any errors or missing values, and transforming the data into a format that is easy to work with.
2. **Exploratory data visualization:** Once your data is clean and prepared, you can start to explore it using data visualization techniques. This can help you to identify patterns, trends, and outliers in your data.
3. **Statistical summaries:** In addition to data visualization, you can also use statistical summaries to explore your data. This can help you to understand the distribution of the data, and to identify any relationships between different variables.
4. **Hypothesis testing:** If you have any specific questions about your data, you can use hypothesis testing to test your assumptions.
5. **Modeling:** Once you have a good understanding of your data, you can start to build models to make predictions or to understand the relationships between different variables.

EDA

The follows are some examples of EDA:

- **Univariate analysis:** This is the simplest form of EDA, where you analyze one variable at a time. You can use statistical summaries, such as the mean, median, and standard deviation, to understand the distribution of the variable. You can also use data visualization techniques, such as histograms and box plots, to visualize the distribution of the variable.
- **Bivariate analysis:** This is where you analyze two variables at a time. You can use statistical techniques, such as correlation and regression, to understand the relationship between the two variables. You can also use data visualization techniques, such as scatter plots and line charts, to visualize the relationship between the two variables.
- **Multivariate analysis:** This is where you analyze three or more variables at a time. You can use statistical techniques, such as multivariate regression and principal component analysis, to understand the relationships between the multiple variables. You can also use data visualization techniques, such as heatmaps and cluster analysis, to visualize the relationships between the multiple variables.

Modeling

- **Modeling** is the process of creating a simplified representation of a system or process.
- **Models** represent the system or process using a variety of techniques, such as mathematical equations, computer simulations, or physical model.
- **Model fitting** is an iterative process as there are typically multiple modeling approaches that are applicable for a given task.
- Even for a single modeling approach, there is a list of parameters which we can set to obtain different configuration of the model.

Note:

- **Model parameters** are estimated from data automatically by algorithms, e.g., weights
- **Model hyperparameters** are set manually and are used in processes to help estimate model parameters, e.g, k in KNN, learning rate, number of layers in NN, number of epochs, loss function, etc.

Data Modeling

Data modeling is a more general process than data fitting.

- **Data modeling** is the process of creating a conceptual representation of data.
- Data modeling can be used to create a variety of different types of models, including mathematical models, statistical models, and conceptual models.
- It is used to understand the data, to communicate the data to others, and to build systems that use the data
- E.g., ERD (data entities and relationships), DFD, UML, etc.

Data Fitting

Data fitting is specifically focused on finding mathematical functions that best describe an existing data.

- It is used to interpolate/extrapolate data points in a dataset.
- Example techniques are least squares regression, polynomial fitting, spline interpolation.

Model Fitting

- **Emphasis on the model:** Emphasize on **mathematical or statistical model** itself. The objective is to adjust the *parameters* of a pre-selected model (which represents a theoretical relationship or hypothesis) so that this model can best explain or predict the observed data.
- **Used** in statistics, machine learning, and econometrics, where models (like linear regression, neural networks, decision trees) are built to understand relationships, make predictions, or classify outcomes.
- **Focus on generalization:** A key concern is not just how well the model fits the *training* data, but how well it *generalizes* to new, unseen data.
 - ✓ “Underfitting” (model too simple)
 - ✓ “Overfitting” (model too complex and learns noise)
- **Example techniques:** gradient descent, maximum likelihood estimation, various optimization algorithms used to train machine learning models.

ML methods used in DS

Supervised Learning

- Classification
 - Predictive method
 - Naïve Bayes
 - K-nearest neighbors (k-NN)
 - Support vector machine (SVM)
 - Decision trees
 - Random forest
 - Logistic regression
- Regression
 - Predictive method
 - Linear regression

Unsupervised Learning

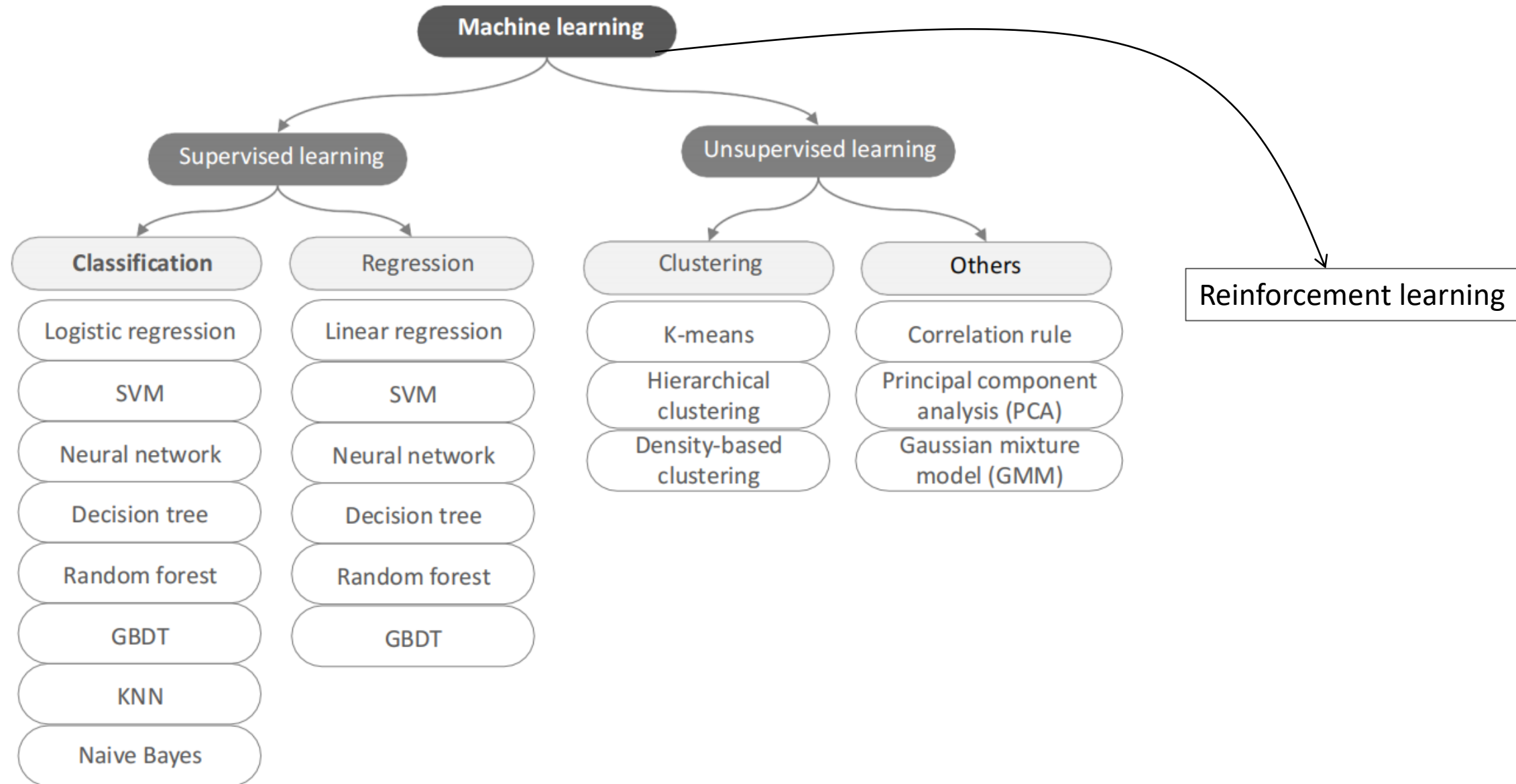
- Clustering
 - Descriptive method
 - Density-based spatial clustering of applications with noise (DBSCAN)
 - K-means
 - Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH)

- Visualization
 - Descriptive (analytics), e.g., box plots, scatter plots, etc.

Association rules

- Descriptive method
 - Apriori (e.g., market basket analysis)
- Anomaly detection (or outlier detection)
- Descriptive

ML algorithms (another way to look at)



Data mining (DM)

The models produced by the DM step fall under two types:

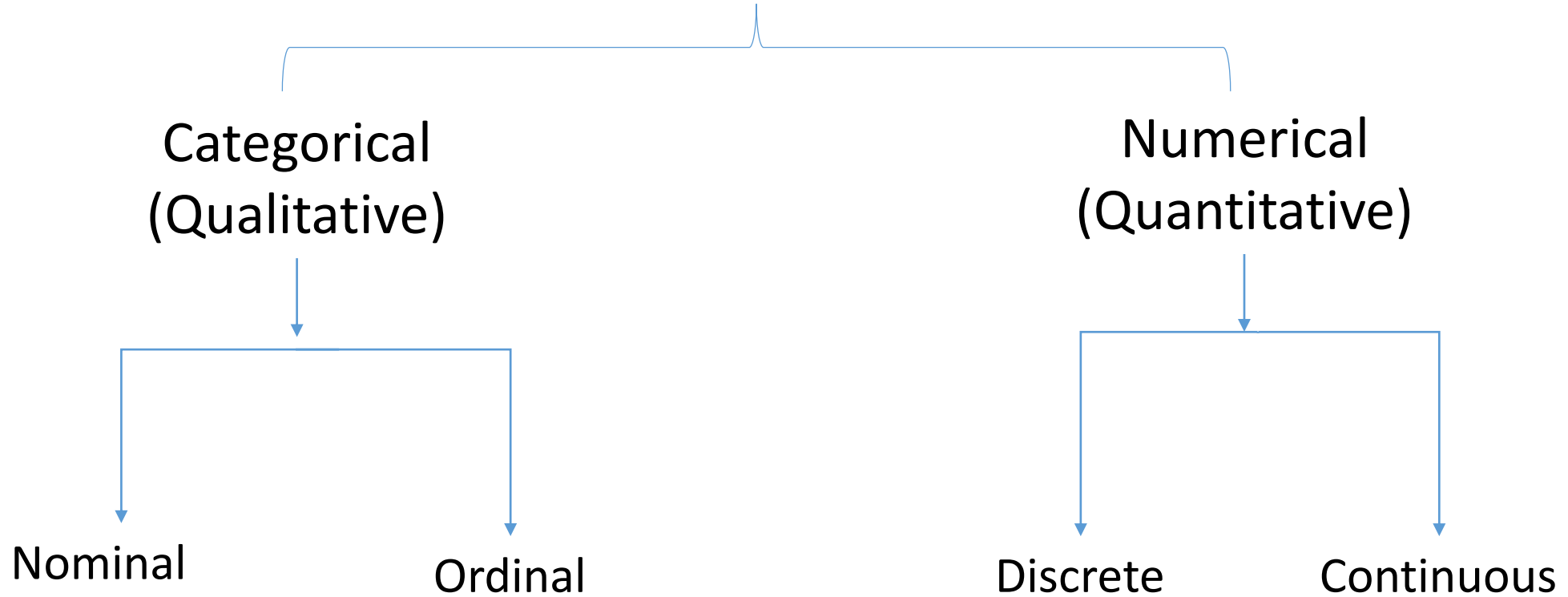
1. Predictive model

- the goal is to predict the value of a variable given the remainder of the variables. The variable to be predicted is called the dependent or *target* variable, and the variables used for prediction are called independent or *predictor* variables.

2. Descriptive model

- the goal is to identify the relationships between variables to learn more about the structure of the data

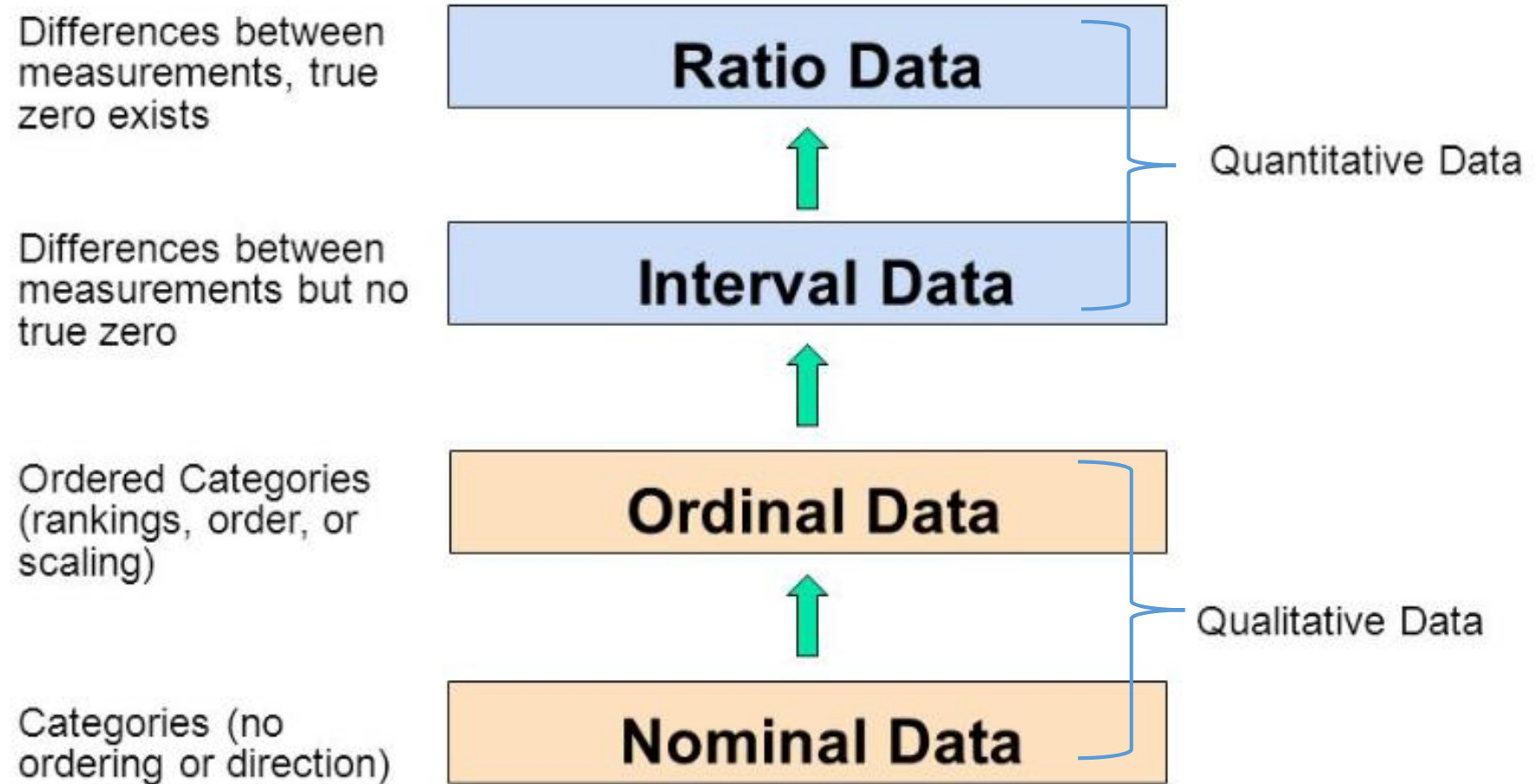
Types of data/variables



Types of data (cont'd)

- Numerical data (also known as quantitative)
 - Discrete
 - Distinct numbers with step-size values
 - E.g., population, number of coins in a pocket
 - Continuous
 - Not limited by step size; not limited by number of positions after decimal point, but usually be rounded off to some positions
 - E.g., amount of water in a cup; real value from 0 to 1, weight, height
- Categorical data (also known as qualitative)
 - Nominal data
 - It describes characteristics; we can assign numbers to it but cannot mathematically compare them, e.g., gender = { male, female, unspecified}; races = { white, black, Asian, Hispanic}
 - Ordinal data
 - Some kind of mixed nature of both numerical and categorical data
 - Can be compared, but not as straightforward; need many data points to meaningfully compare, e.g., average rating stars of the hotels; 5-star rating = {*, **, ***, ****, *****} of the hotels

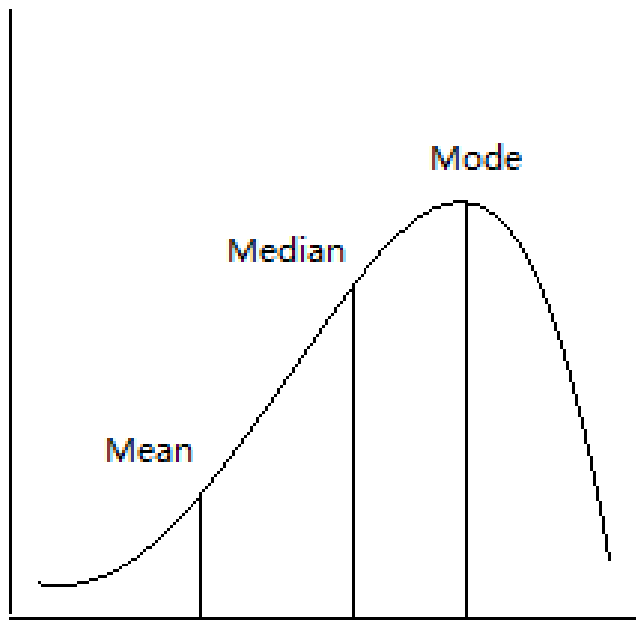
Scales of measurement



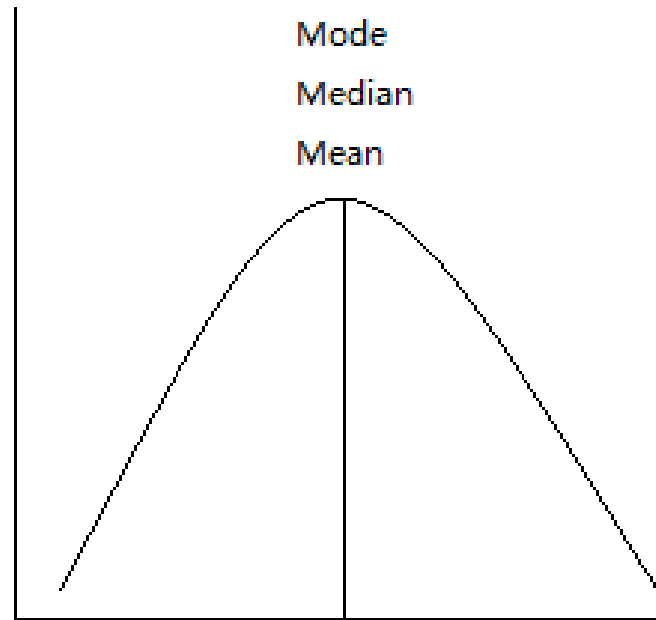
Examples of scales of measurement

- 1) **Nominal** variables include: blood type, zip/area code, gender, race, eye color, political party, etc.
- 2) **Ordinal** variables include: socio economic status (low income, middle income, high income), education level (high school, BS, MS, PhD), satisfaction rating (extremely dislike, dislike, neutral, like, extremely like)
- 3) **Interval** variables include: temperature (Fahrenheit), temperature (Celsius), pH, TOEFL score (310-677), standardized exam score, AD year ('in the year of the Lord'), BE year (Buddhist Era), etc.
- 4) **Ratio** variables include: dose amount, flow rate, concentration, pulse, weight, length, temperature (in Kelvin where 0.0 Kelvin really does mean “no heat” = -273.15 Celsius), etc.

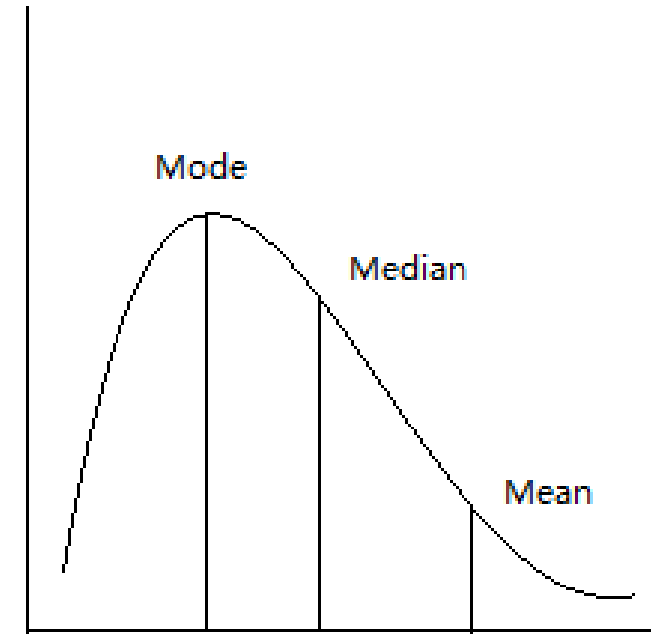
Three types of average



Left skew



Normal Distribution



Right skew

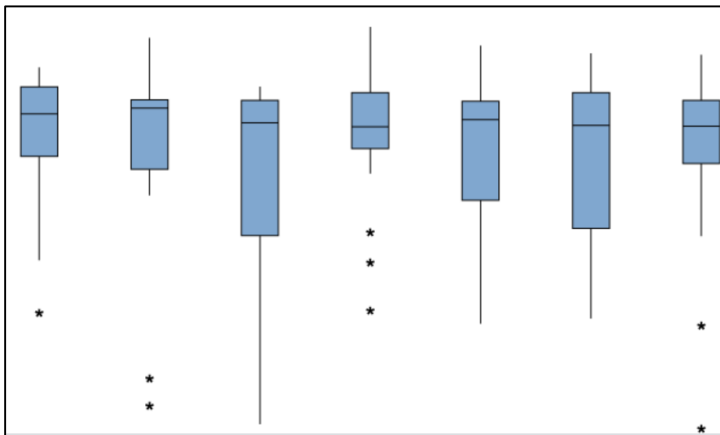
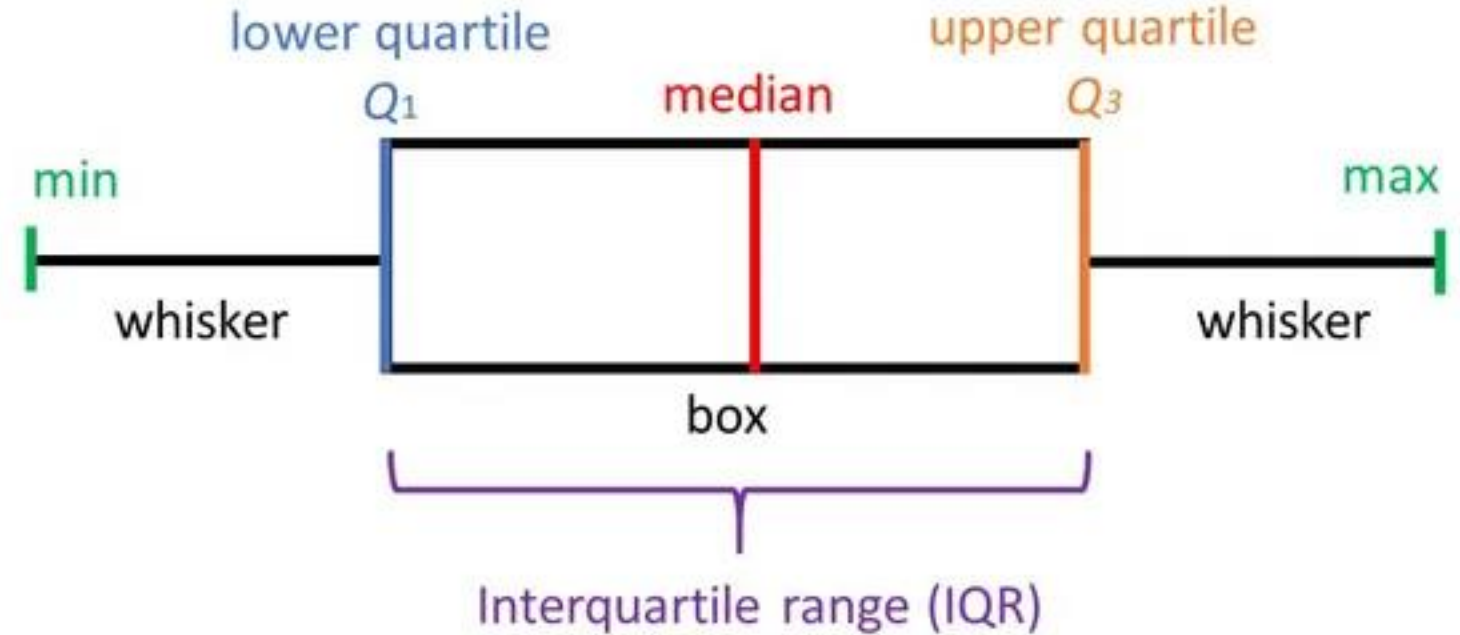
Three types of average

- Mean
 - Typical “average” value as we know it; easy to understand
 - Take into account all the data
 - Affected by outliers
 - E.g., exam score of a class, time to drive back home on average
- Median
 - Middle value of data (after being sorted)
 - Doesn't tell much about the rest of the data
 - E.g., household income of a country
- Mode
 - The most common value of the data
 - Peak of a histogram
 - Doesn't tell much about the rest of the data
 - E.g., employees' income in a company

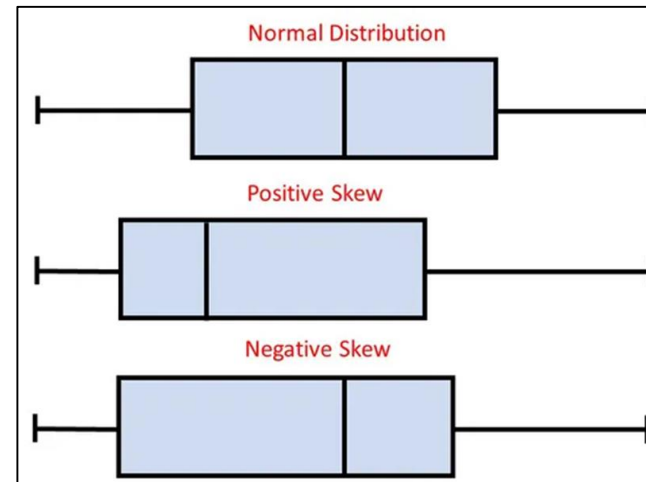
There is no single best measure (either Mean, Median, or Mode), but using just one is a bad idea.

Box plot (also known as box and whisker plot)

- Show dispersion (also called variability, scatter, or spread) of the data
- Show skewness, if any
- Show quartiles
- Show outliers, if any
- Can be used for comparison when having different groups



Credit: <https://www.simplypsychology.org/boxplots.html>



Note:

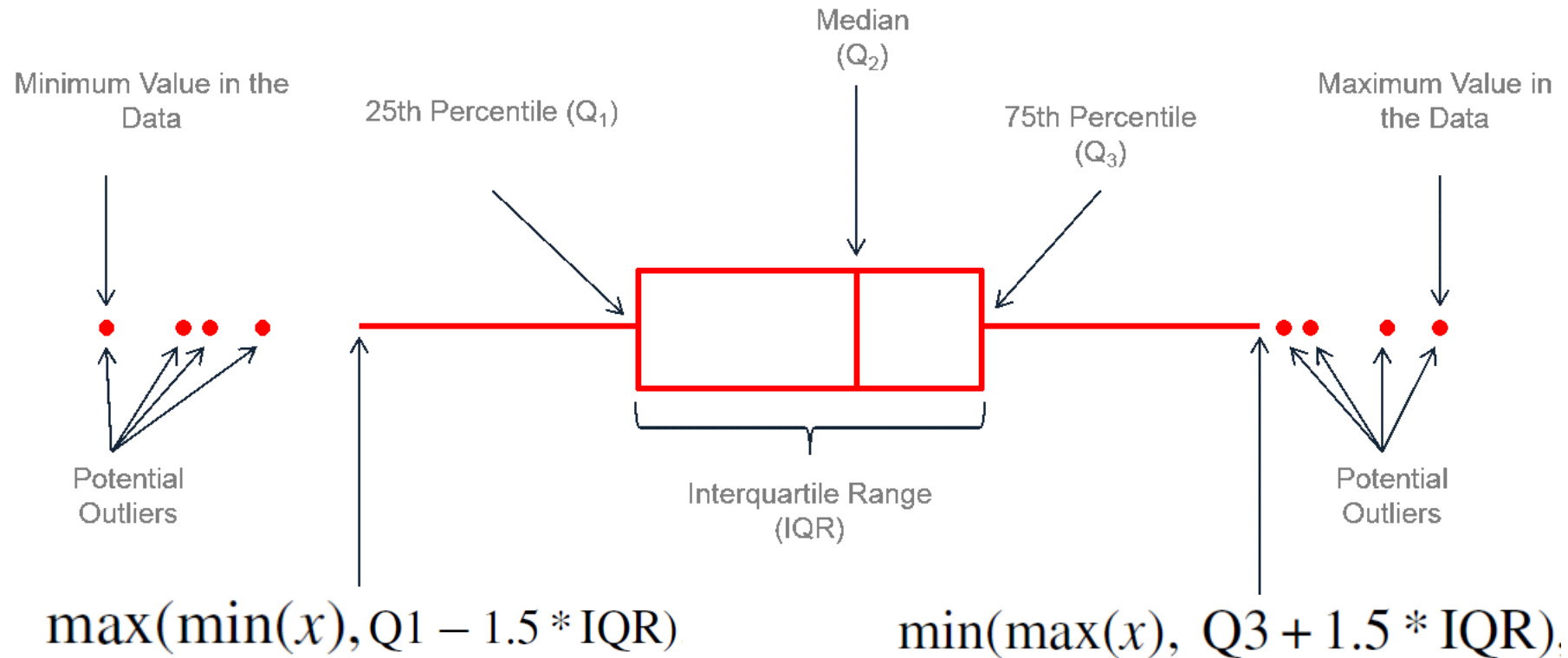
- Because the whisker must end at an observed data point, the whisker lengths can be unequal.
- Two whiskers are drawn to the smallest or largest observed data points from the dataset that falls within the range of $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.

Terms and notations in the boxplot

A boxplot is a standardized way of displaying the dataset based on the **five-number summary**: the minimum, the maximum, the sample median, and the first and third quartiles.

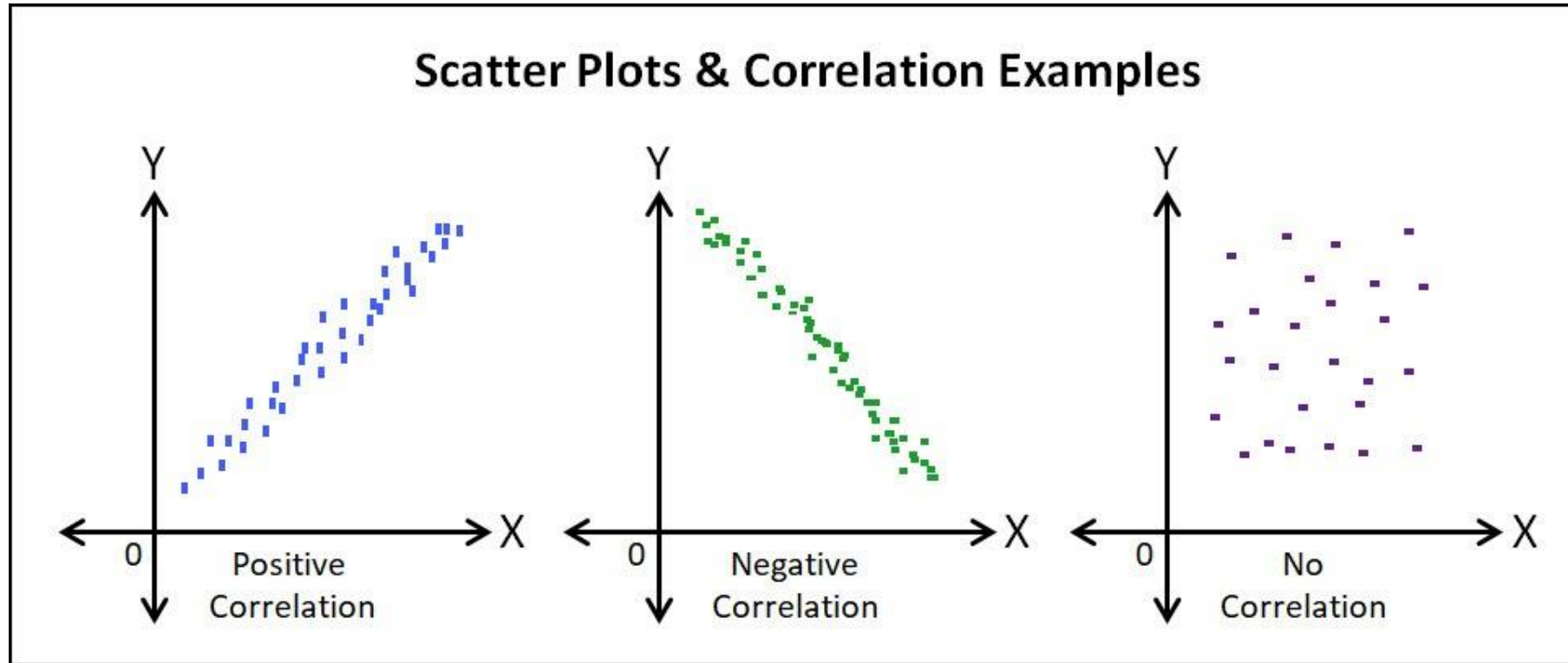
- **Minimum** (): the lowest data point in the data set excluding any outliers
- **Maximum** (): the highest data point in the data set excluding any outliers
- **Median** (Q_2 or **50th percentile**): the middle value in the data set
- **First quartile** (Q_1 or **25th percentile**): also known as the *lower quartile* $q_n(0.25)$, it is the median of the lower half of the dataset.
- **Third quartile** (Q_3 or **75th percentile**): also known as the *upper quartile* $q_n(0.75)$, it is the median of the upper half of the dataset.^[7]

Box plot (also known as box and whisker plot)



Note: Values of **x** must fall between the **fences**, not including the outlier values.

Scatter plot



- Show every data point
- Show how dense or sparse the data is; show spread
- Show correlations and trends, if any

Correlation (สหสัมพันธ์)

Coefficient of Correlation

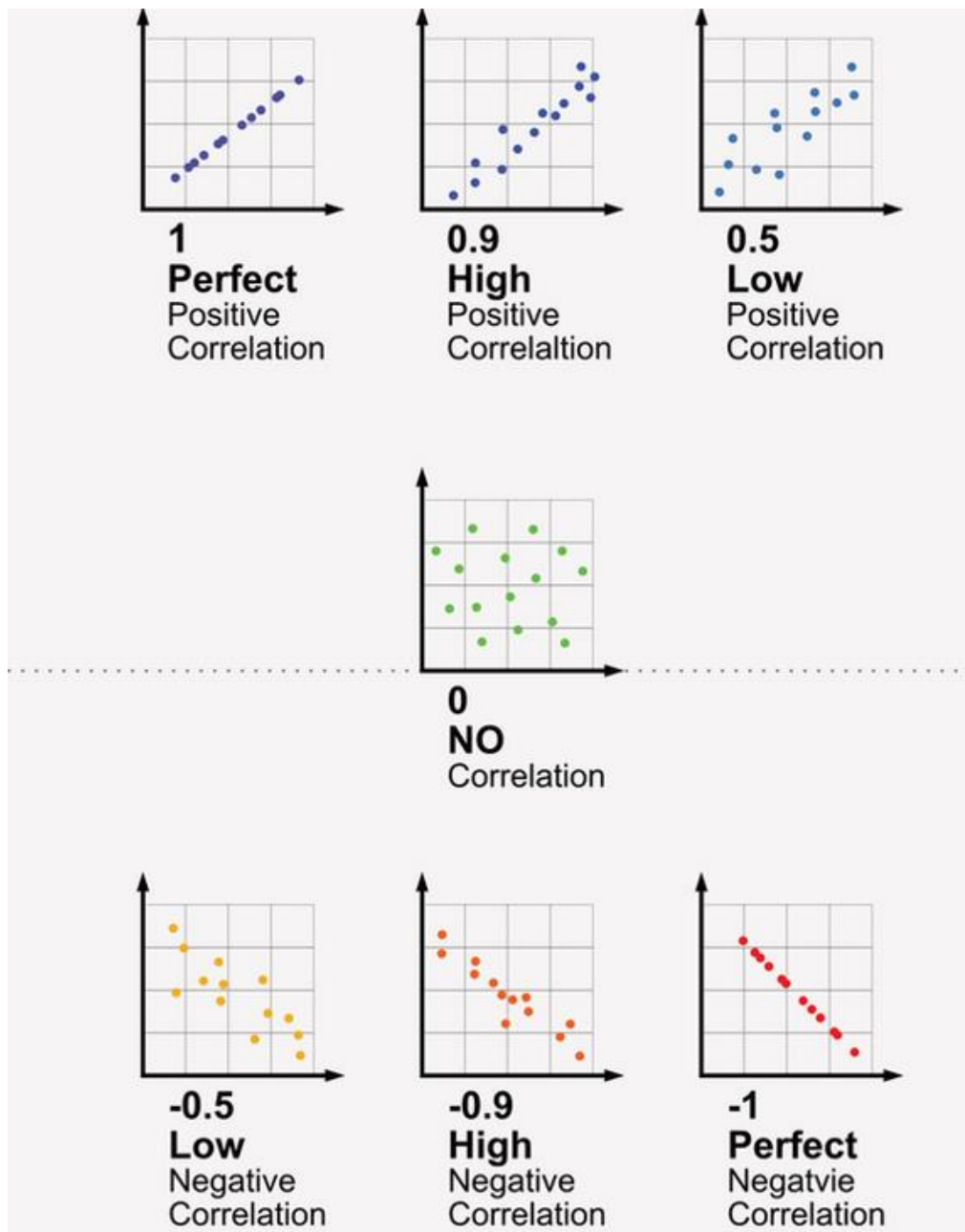
$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$-1 \leq r \leq 1$$

Note:

Pearson Correlation → for numerical data;
only for linear relationship

Spearman Correlation → for ordinal scale (ranked order);
for monotonic/linear relationships



Linear modeling

- Try to answer the question
 - Does x influence y ?
 - Are house prices influenced by incomes?
 - Is changing the oil of a car more frequently a good thing? Does it save money in the long run?
- Linear modeling, also known as linear regression
- Statisticians have an old saying that is very important to remember:
“Correlation does not mean causation.”
- The idea that for many kinds of data it is possible to fit a line to a set of data points, which represents the connection between an independent variable and a dependent variable

Linear regression

- The target variable is numerical, e.g., inches of rain for the day, and the predictor variables can be numerical, categorical, or ordinal
- Whereas in classification, the linear regression is used to predict a continuous value, but when the target variable is categorical, e.g., weather forecast for the day (sunny, cloudy, rain, snow), some threshold values can be used (e.g., quantization) to classify.
- Generally, logistic regression (based on probability) is more suitable for categorical classification. (We'll learn more about it soon later.)

Linear regression

Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Parameters:

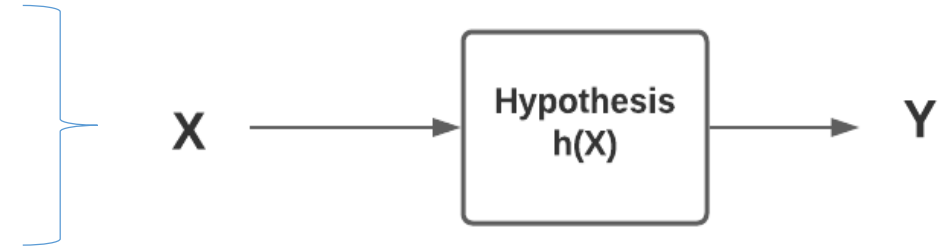
$$\theta_0, \theta_1$$

Cost Function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

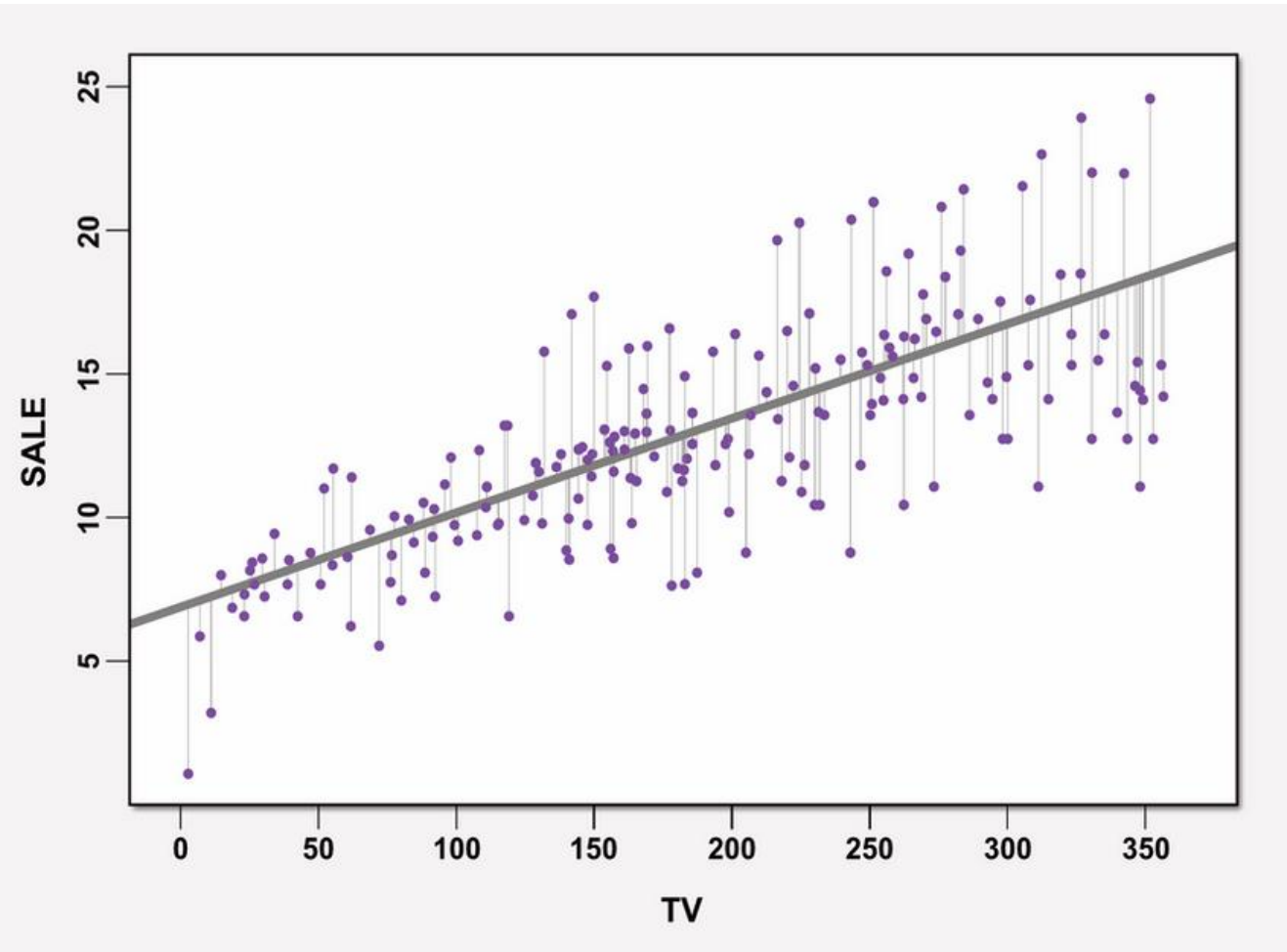
Goal:

$$\underset{\theta_0, \theta_1}{\text{minimize}} J(\theta_0, \theta_1)$$



For regression model, the most popular loss function (cost function) used is **least squares method**

Simple linear regression (การวิเคราะห์การถดถอย)



$$y = ax + b$$

a = Slope
 b = Y-Intercept

Linear model

$$y = f(x_1, \dots, x_m; w)$$

$\mathbf{x}$ is independent variables, y is target variable, $f()$ is a linear function,
 $\mathbf{w}$ is parameter set

Two steps

- **Model Fitting:** For a given dataset with variables $x_1, \dots, x_m$ and y , we calculate the model parameters w that best meet certain criteria.
- **Model Prediction:** Given a dataset with the predictor variables $x_1, \dots, x_m$ and model parameters w , we calculate the value of the target variable y .

Simple linear regression

Here we focus on one input variable and one target variable.

$$y = f(x; w) = w_0 + w_1x + \varepsilon$$

w is intercept and slope, ε is error or residual

$$\varepsilon_i = y_i - w_0 - w_1x_i$$

Simple linear regression

Loss function

$$w^* = \arg \min_w \sum_i \varepsilon_i^2 = \arg \min_{w_0, w_1} \sum_i (y_i - w_0 - w_1 x_i)^2.$$

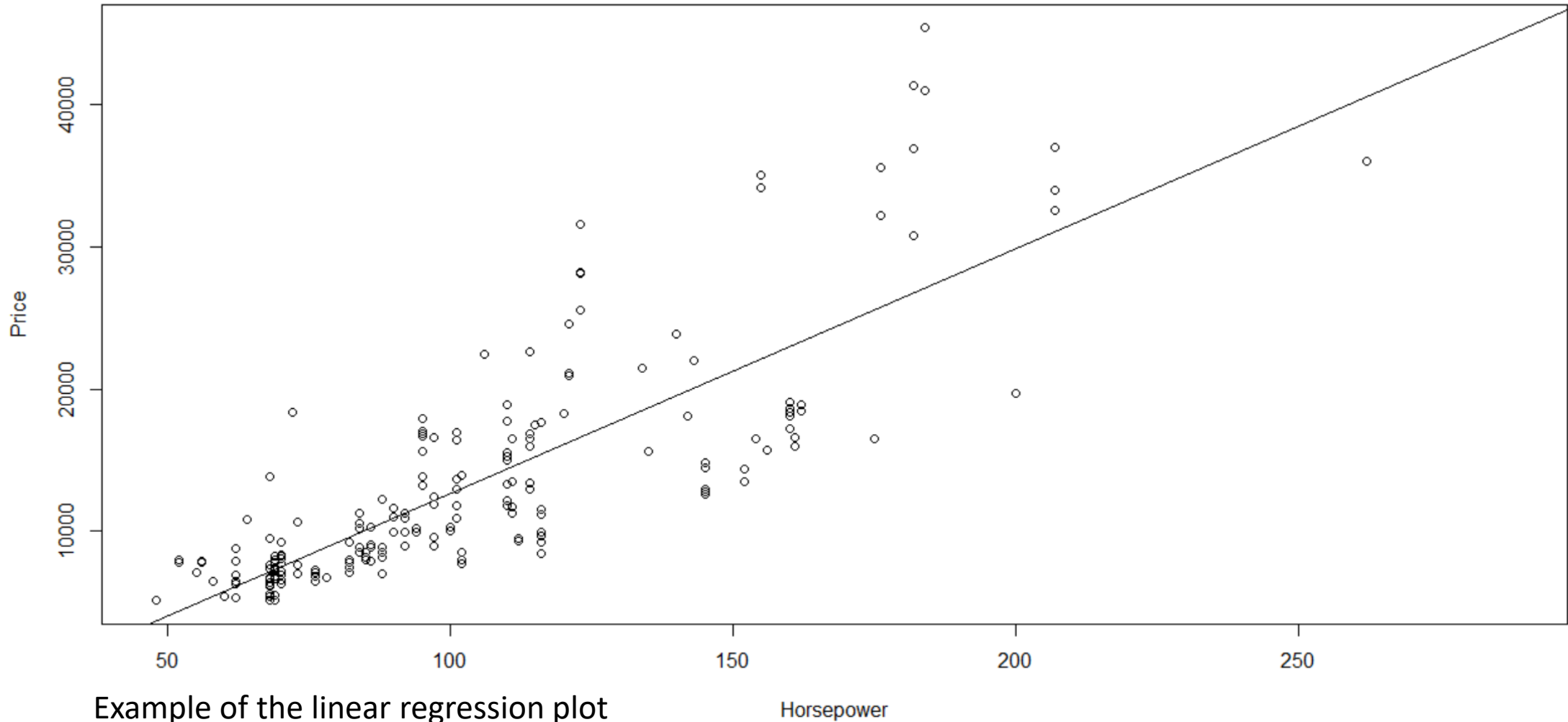
Note: The **arg min** function returns the argument or arguments (arg) for the target function that returns the minimum (min) value from the target function.

Take derivative of the loss function and set to zero, and solve for

$$w_1^* = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2},$$

$$w_0^* = \bar{y} - w_1^* \bar{x},$$

Linear regression on numerical variables



Try to work on the UCI's automobile dataset; see our class assignment

Multivariate linear regression

(also-called **multiple linear regression**)

- We want to look at how multiple input variables have effect on the target variable.
- We are treating the target variable as a function of the different predictor variables.
- We can observe the effect of changing one predictor variable while controlling the values of the other predictor variables.

$$y = w_0 + w_1x_1 + \cdots + w_mx_m = w_0 + \sum_{i=1}^m w_ix_i$$

Here we have

$m + 1$ model coefficients $w_0, \dots, w_m$.

Example (from our in-class example code)

```
call:
lm(formula = auto$Price ~ auto$Horsepower + auto$EngineSize)
```

Residuals:

Min	1Q	Median	3Q	Max
-11180	-1908	-201	1629	13277

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8510.47	843.84	-10.085	< 2e-16 ***
auto\$Horsepower	60.21	12.24	4.917	1.85e-06 ***
auto\$EngineSize	122.44	11.01	11.118	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3678 on 196 degrees of freedom

(6 observations deleted due to missingness)

Multiple R-squared: 0.7896, Adjusted R-squared: 0.7875

F-statistic: 367.8 on 2 and 196 DF, p-value: < 2.2e-16

- A **t-test** will tell you if a single variable is statistically significant.
- An **F-test** will tell you if a group of variables are jointly significant (the larger, the better). Also we would use the p-value to get the “big picture.”

More on F-test

The F-test is often used in ANOVA (analysis of variance) to compare the variances of the different groups in the study. For example, if you are comparing the average test scores of students in different classrooms, you could use an F-test to see if the variances of the test scores are different between the classrooms.

The F-test is also used in regression analysis to test the significance of the model. The F-test statistic is the ratio of the variance explained by the model to the residual variance. A large F-test statistic indicates that the model is significant and that the independent variables are able to explain a significant amount of the variance in the dependent variable.

Example (from our in-class example code)

```
Call:
lm(formula = auto$Price ~ auto$Horsepower + auto$Length + auto$EngineSize)

Residuals:
    Min       1Q   Median       3Q      Max
-9227.5 -1928.1  -14.2   1313.1 12984.0

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -24836.65    4232.55  -5.868 1.87e-08 ***
auto$Horsepower    58.35     11.83   4.934 1.72e-06 ***
auto$Length    110.13     28.02   3.931 0.000118 ***
auto$EngineSize  101.45     11.89   8.529 4.05e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3550 on 195 degrees of freedom
(6 observations deleted due to missingness)
Multiple R-squared:  0.8051,    Adjusted R-squared:  0.8021
```

So we get:

$$\text{Price} = 58.35 * \text{Horsepower} + 110.13 * \text{Length} + 101.45 * \text{EngineSize} - 24,836.65$$

More example with categorical variables

```
call:
lm(formula = auto$Price ~ auto$Horsepower + auto$EngineSize +
    auto$Brand)
```

Residuals:

Min	1Q	Median	3Q	Max
-4878.2	-1162.8	-71.3	995.2	10015.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-408.062	1580.807	-0.258	0.79660	
auto\$Horsepower	64.381	9.394	6.853	1.17e-10	***
auto\$EngineSize	57.068	10.223	5.582	8.86e-08	***
auto\$Brandaudi	3438.746	1590.140	2.163	0.03193	*
auto\$Brandbmw	8062.712	1536.941	5.246	4.43e-07	***
auto\$Brandchevrolet	-2203.927	1877.326	-1.174	0.24199	
auto\$Branddodge	-3133.680	1517.878	-2.065	0.04044	*
auto\$Brandhonda	-2239.854	1465.799	-1.528	0.12829	
auto\$Brandisuzu	-2646.239	2067.272	-1.280	0.20221	
auto\$Brandjaguar	5814.388	2095.089	2.775	0.00611	**
auto\$Brandmazda	-323.508	1428.641	-0.226	0.82112	
auto\$Brandmercedes-benz	11713.483	1705.314	6.869	1.07e-10	***

Use only one brand in a prediction at a time, by multiplying 1.0 to its chosen brand's coefficients while multiplying 0.0 to the others.

So we may get a predicted pricing for a BMW brand car as:

$$\text{Price} = 64.38 * \text{Horsepower} + 57.07 * \text{EngineSize} + 1.0 * 8,062.71 - 408.06$$

Meaning of some coefficients and values

- **R-Square หรือ Coefficient of determination** เป็นค่าที่ใช้พิสูจน์ว่า Model ที่ได้นั้นเหมาะสมหรือไม่ โดยค่า R-Square จะอยู่ระหว่าง 0-1 ซึ่งยิ่งเข้าใกล้ 1 ยิ่งดี และโดยทั่วไปควรมีค่า 0.6 ขึ้นไป และจะดีมากเมื่อมีค่ามากกว่า 0.8 ขึ้นไป แต่ก็ไม่ได้มีกฎเกณฑ์แน่นอนตายตัวแล้วแต่กรณี
- **Adjusted R-Square** เป็นค่าที่บ่งบอกว่า R-Square ที่ได้นั้นเหมาะสมหรือไม่ โดยการลดจำนวนชุดข้อมูลลง 1 ชุด แล้วคำนวณ R-Square ใหม่ ซึ่งหากมีค่าต่ำกว่า R-Square มากผิดปกติก็สามารถสรุปได้ว่าใช้ชุดข้อมูลน้อยเกินไป หรือ โมเดลนั้นมีผลสืบเนื่องจากการเปลี่ยนแปลงจำนวนชุดข้อมูลเป็นอย่างมาก มีโอกาสที่ Model จะผิดพลาดสูง ค่า Adjusted R-Square สำหรับโมเดลที่ดีจะมีค่าต่ำกว่า R-Square เล็กน้อย

Note: R^2 shows how well terms (data points) fit a curve or line. If you add more and more useless variables to a model, adjusted r-squared will decrease. If you add more useful variables, adjusted r-squared will increase. Adjusted R^2 will always be less than or equal to R^2 .

Meaning of some coefficients and values

- **P-value** คือ ความน่าจะเป็นที่จะได้ผลลัพธ์เท่ากับหรือเกินกว่าที่สังเกตได้ภายใต้สมมติฐานหลัก มีความสำคัญอย่างยิ่งในการทำ Regression เพราะมันเป็นตัวบ่งบอกว่า ตัวแปรนั้นๆ มีความสำคัญต่อระบบความสัมพันธ์หรือสมการหรือไม่ ซึ่งค่า P-value จะอยู่ระหว่าง 0-1 วิธีการอ่าน P-value ง่ายๆ คือ ค่า P-value ที่มากกว่า 0.05 บ่งบอกว่าตัวแปรนั้นอาจจะมีไม่ความสำคัญต่อระบบสมการ ซึ่งเราสามารถลองตัดออกแล้วทำการ Regression ใหม่ เพื่อหาสมการที่เหมาะสมต่อไปได้

Now try it yourself

- As an assignment, work on the smartwatch's heart rate dataset (or pick any publicly available dataset) provided in class.
 - Do box plots, histograms (or distributions), scatter plots
 - Perform linear regression (intercept and coefficient values)
 - Find correlations (r value), R-squared and adjusted R-squared values